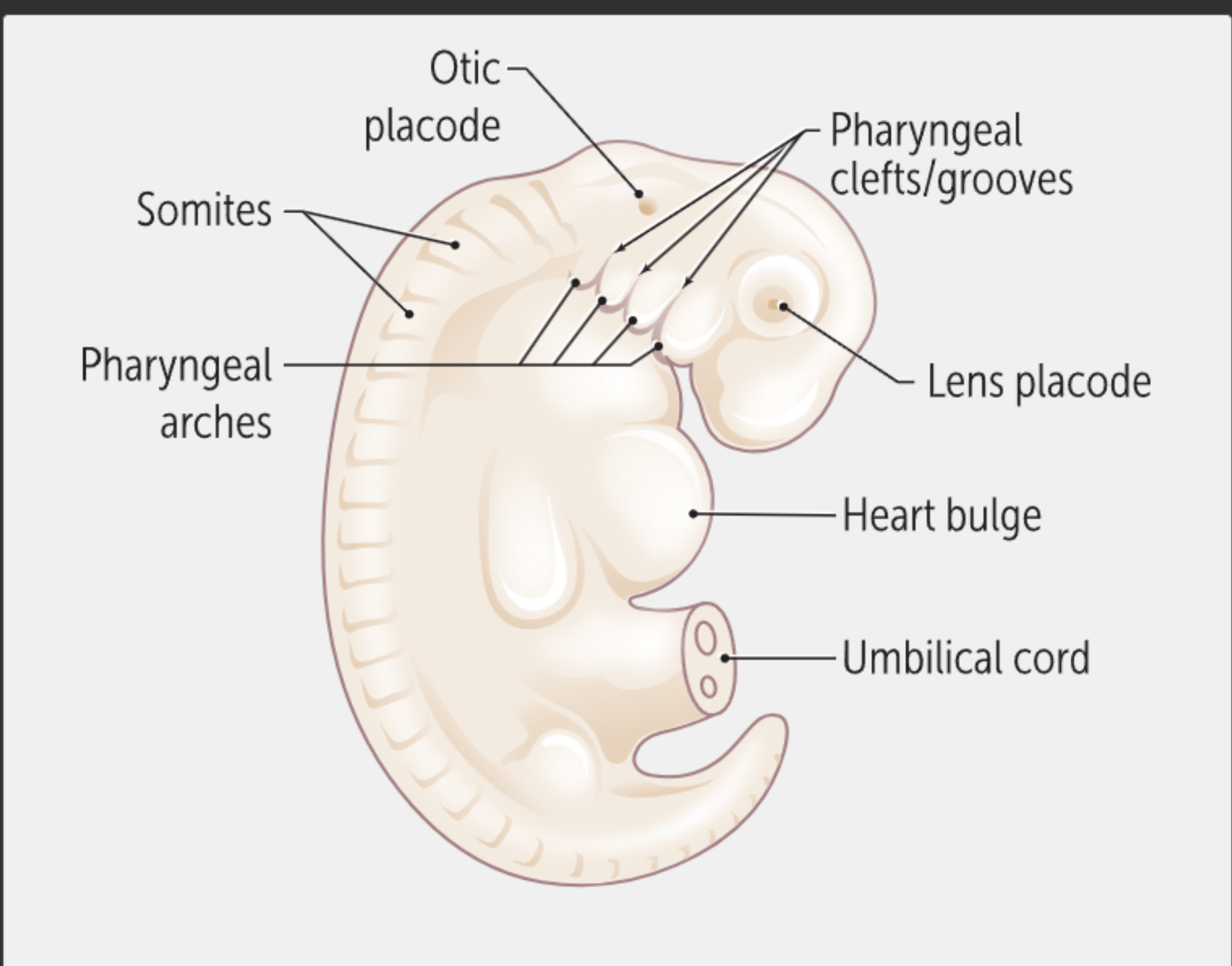


- **Artery** (called an aortic arch)

These embryonic aortic arches should not be confused with the single adult aortic arch, which is only one of their many derivatives.



QUIZ



What do you want to learn?



ALICE L WALTON SCHOOL OF MEDICINE

Management of Shock

Last updated March 3, 2026 19 min read ScholarRx
Adapted by: Diptiman Bose, Phil Riley, Trager Hintze, Robert Donnell
Editors: Jordan Finn, Russell Johnson

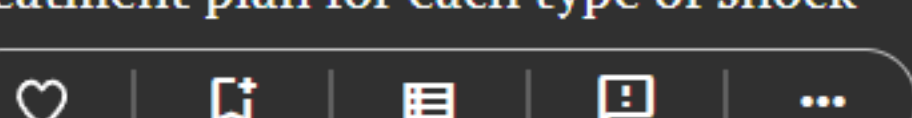
Adapted from: Management of Shock



Learning Objectives (4) Versions

After completing this brick, you will be able to:

- 1 Describe the hemodynamic findings for each type of shock.
- 2 Explain the use of fluid resuscitation in shock.
- 3 Justify the use of pharmacological agents in shock treatment
- 4 Outline the treatment plan for each type of shock



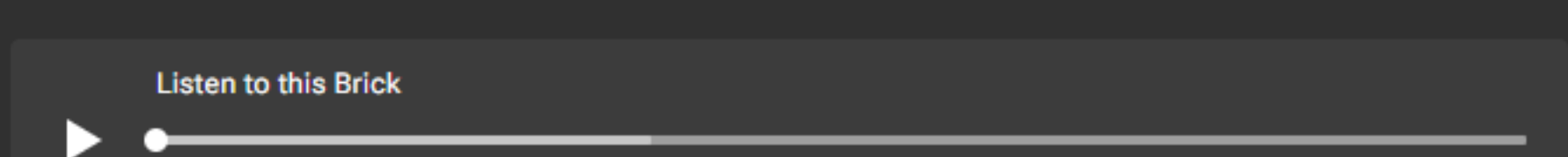
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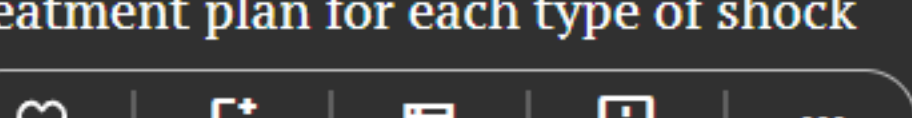
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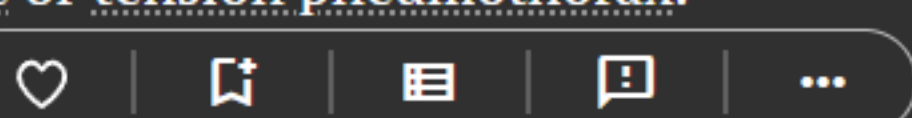


but it can rapidly progress to multiorgan system failure and death if not corrected quickly. This brick will focus on how we stabilize the patient while looking for the underlying cause of shock (eg, infection, trauma, or heart failure).

Types of Shock

The five types of shock are:

- **Distributive shock** arises from systemic vasodilation, which might be triggered by conditions such as sepsis or anaphylaxis.
- **Cardiogenic shock** can occur from a condition such as heart failure or myocardial infarction that leads to low stroke volume (SV) and low cardiac output (CO).
- **Hypovolemic shock** is caused by low intravascular volume that leads to low preload and low CO and can be caused by hemorrhage, GI blood loss, or severe dehydration.
- **Neurogenic shock** is caused by the loss of sympathetic output from the central nervous system (CNS), leading to a global loss of arterial tone, cardiac output, heart rate, and preload (venous tone); neurogenic shock can follow head or spinal cord trauma.
- **Obstructive shock** is due to an obstruction in blood flow through the heart. There are two types. In the pulmonary type, a pulmonary embolus causes increased resistance to flow within the pulmonary circulation and prevents blood from reaching the left side of the heart. In the structural type, there is global obstruction of flow through the heart caused by cardiac tamponade or tension pneumothorax.



Cardiogenic shock: Inotropes are used to treat cardiogenic shock. The two inotropes used in cardiogenic shock are dobutamine and milrinone. As discussed, many adrenergic vasopressors also have strong inotropic activity, most notably epinephrine and dopamine.

CLINICAL CORRELATION

Early assessment of cardiac function is critical in shock evaluation. While formal echocardiography requires a technician, equipment transport, image acquisition, and delayed cardiologist interpretation, Point Of Care UltraSound (POCUS) allows you to assess cardiac function in real-time at the bedside. POCUS reveals the heart's systolic function, measured as ejection fraction (EF). A newly reduced EF in the setting of shock suggests cardiogenic etiology—though this determination requires knowledge of the patient's baseline function. Normal or supranormal EF points toward distributive, hypovolemic, or obstructive shock.

The **pulmonary type of obstructive shock** (from a pulmonary embolus) is treated initially with IV fluids (and pressor drugs if needed), then definitively by dissolving the embolus, often with thrombolytic drugs like tissue plasminogen activators.

The **structural causes of obstructive shock** are treated initially with IV fluids, then with surgical techniques that correct the obstruction:

- Cardiac tamponade is treated by withdrawing pericardial fluid (pericardiocentesis).
- Tension pneumothorax is treated with a thoracostomy tube placed to preferred).
- Bolus strategy: Rapid IV boluses (500-1000mL), reassess, repeat as needed
- Blood products if bleeding: If hemorrhagic shock, transition to packed red blood cells (pRBCs) early.
- Monitor hemodynamic and organ perfusion response

Treatment: Distributive Shock

- Septic Shock (most common): Fluids first: 50mL/kg crystalloid bolus within 3 hours (well-established initial resuscitation)
- Antibiotics: Administered as soon as possible
- Vasopressors: Norepinephrine is first-line when fluids aren't enough
- Source control: Find and fix the infection (drain abscess, remove infected catheter, etc.)

Treatment: Anaphylactic Shock

- Epinephrine: first-line (EpiPen) to reverse vasodilation (α -1 mediated vasoconstriction will reverse) and bronchospasm (β 2 mediated bronchodilation will reverse)
- Fluids: Aggressive IV crystalloids
- Adjuncts: H1/H2 blockers, corticosteroids, albuterol if bronchospasm

What class of drugs is used to stabilize blood flow and organ perfusion, except in cases of pulmonary edema?

- Distributive shock is managed with vasopressors like norepinephrine, which cause vasoconstriction by stimulating adrenergic receptors.
- Cardiogenic shock is treated with inotropes such as dobutamine and milrinone, which increase cardiac contractility and may cause vasodilation.
- Obstructive shock is treated by addressing the underlying cause, such as thrombolytics for pulmonary embolism or surgical interventions for cardiac tamponade and tension pneumothorax.

Review Questions

1. A 65-year-old female presents with a 2-week history of cough, fever, and chills. She is confused and disoriented. Her blood pressure is 70/50 mm Hg, pulse is 125 beats/min, and temperature is 102.5°F. Exam shows right flank tenderness, and her urinalysis shows numerous white blood cells and bacteria. The white blood cell count is elevated. She is admitted to the intensive care unit. Her blood pressure does not change with 1 L of normal saline. Which of the following medications would be the best next step in management?

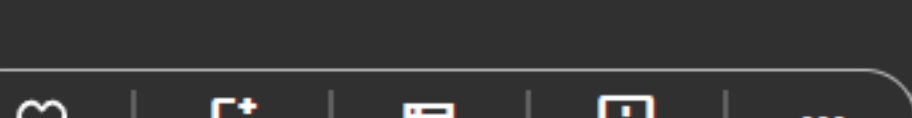
- A. ☐ Digoxin
- B. ☐ Dobutamine
- C. ☐ Milrinone
- D. ☒ Norepinephrine



The correct answer is PDE3 inhibition (D). Milrinone is an inotrope that is used for cardiogenic shock. It works by inhibiting PDE3, leading to increases cAMP and intracellular Ca^{2+} ; the increases in these substances improve cardiac contractility. Milrinone also causes vasodilation. Phenylephrine, epinephrine, and norepinephrine are vasopressors that all induce α 1-adrenergic stimulation (A). Dobutamine causes β 1-adrenergic stimulation (B) to increase heart rate and contractility. Epinephrine, norepinephrine, dobutamine, and dopamine all cause differing degrees of β 2-adrenergic stimulation (C).

3. A 35-year-old female who is 16 weeks pregnant is admitted with confusion and cold extremities. Her blood pressure is 85/75 mm Hg, pulse 110 beats/min, and temperature is 100.4°F. The heart, lung, and abdominal/pelvic exams are normal. Which of the following would be the most appropriate initial treatment?

- A. ☒ Give normal saline
- B. ☐ Start epinephrine
- C. ☐ Start milrinone
- D. ☐ Start norepinephrine
- E. ☐ Start vasopressin



Hide Explanation